Checkpoint #3: Austin

Role D. Circulation · COSC 3380 · Group 6 · Due Tuesday, October 27, 2026

Your schedule

Week	What you do
Week 1 (Oct 8 to 14)	Midterms week, so no coding is expected. Read the loan rules and write the checkout, return and renew steps in plain English (one page).
Week 2 (Oct 15 to 21)	Start Thu Oct 15, with Josh pairing on the first flow. Checkout, return, renew, open loans page. The open-loans query. The loan trigger with Josh.
Week 3 (Oct 22 to 26)	Merge Oct 22, test the full cycle, fix bugs only after the Oct 24 freeze.

What you build

You build the heart of the library: lending an item, taking it back, and extending a loan. It is the busiest area, so the work starts Oct 15 and Josh pairs with you on the first flow.

Tables: Loan, Copy, Member, MemberType, Hold, and Fine (to check unpaid fines).

Build, in this order:

1. **Checkout form (staff only).** Staff enter a copy's barcode and a member. The server checks the rules before saving: the member is active, is under the total and media loan limits for their member type, has no unpaid fines above $2.50, and the copy is available and not reserved for someone else's Ready hold. If the borrower is a Staff employee, a different staff member must process it.
2. **Pick the loan type.** If only 1 copy is available, it is a short loan (3 days, no renewal). With 2 or more, it is a standard loan (14 days). Save the Loan row with the staff member's id.
3. **Return form.** Record `Return_date` and `Return_staff_id` together. The database requires both or neither.
4. **Renew form.** Standard loans only, within the member type's renewal limit, and not if another member has an Active or Ready hold on the item. A renewal adds 1 to `Renewal_count` and moves the due date out 14 days.
5. **Open loans page.** Lists what a member has out and when it is due.

Trigger: when a loan is opened, the copy's `Status` becomes on loan. When it is returned, it goes back to available. Josh pairs with you on this.

Query: open loans for a member.

Watch out for: the database allows only one open loan per copy and checks the due-date windows; never write to `Open_copy_id`, because MySQL fills it in; media means videogames, DVDs, music and board games, and books, newspapers and papers are not media.

Done when: a full cycle of checkout, renew and return works on the sample data, and each rule above is refused with a clear message when it is broken.

Key dates (all teams)

Date	What
Wed Oct 14	Skeleton live on Azure, login working
Wed Oct 21	Every area's pages merged and deployed
Sat Oct 24	Code freeze: bug fixes only
Mon Oct 26	Final test as each role, email sent to the TAs and uramamur@bcm.edu
Tue Oct 27	Checkpoint #3 deadline (safety day)

Working rules

- Stack: Node.js, Express, the mysql2 driver, plain HTML/CSS/JavaScript, MySQL.
- Test on your own local MySQL copy of the project dump, not on the hosted database.
- Keep database passwords out of GitHub. They go in a local `.env` file that is never committed.
- Merge into main at least every two days. One person owns each file.
- Deactivate or withdraw records instead of deleting them. History tables keep their rows.
- Stuck for more than an hour? Say so in the group chat.